

ASUMBI GIRLS' 2025 PREDICTION



CONFIDENTIAL KCSE EXAM



231/2

BIOLOGY

PAPER 2 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL.....

SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

1. Write your name and Admission Number in the spaces provided above.
2. Sign and write the date of examination in the spaces provided above.
3. This paper consists of two sections: **A** and **B**. Answer **ALL** questions in section **A** in the spaces provided. In section **B**, answer question 6 (**compulsory**) and either question 7 or 8 in the spaces provided after question 8.

For Examiners' use only

Section	Questions	Maximum Score	Candidate's Score
A	1 - 5	40	
B	2	20	
	3	20	
	4		
Total Score		80	

SECTION A (40 MARKS)

Answer ALL the questions in this section in the spaces provided.

1. Pure breed of red cows and pure breed of white bulls were crossed to give F₁ calves which had a mixture of red and white coat known as roan. The F₁ were selfed.

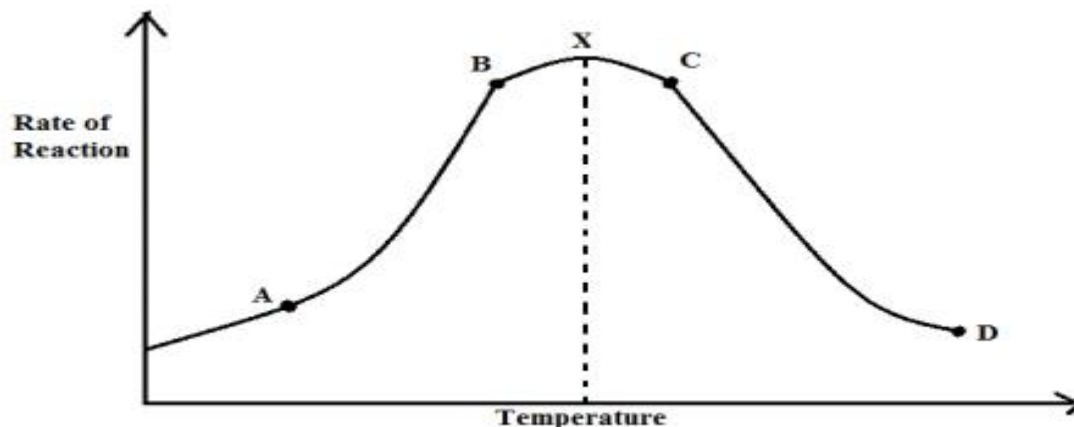
(a) Using letter R to represent gene for red colour and W to represent gene for white colour work out the phenotypic ratio of F₂. **(4 marks)**

(b) Work out the genotypic ratio of a cross between F₁ offspring and white bull. **(3 marks)**

(c) Comment on the gene(s) controlling the colour of coats in cattle mentioned above.

(1 marks)

2. The graph below show the effect of temperature on an enzyme catalysed reaction.



(a) Account for the shape of the curve between:

(i) A and B (3 marks)

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(ii) C and D (2 marks)

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(b) What does the point marked X represent? (1 mark)

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(c) Apart from temperature. State two other factors that affect the rate of enzyme-controlled reaction. (2 marks)

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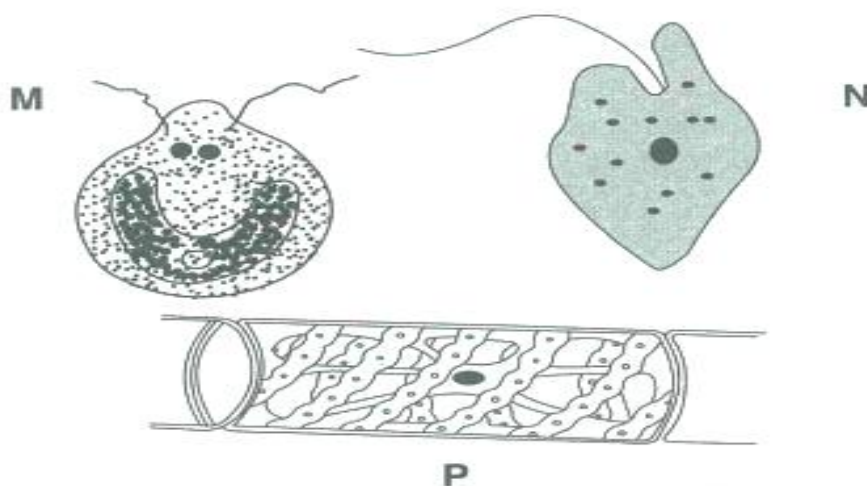
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3. (a) The scientific name *Lantana camara* refers to a green herbaceous plant. Other related plants include *Lantana trifoliata* and *vitex rifoliata*.

(i) From the list, identify the plants belonging to the same genus. (1 mark)

(ii) From the name *Lantana camara*, which name represents the genus name? (1 mark)

(b) During a taxonomical study of pond water, pupils observed the following organisms:



(i) Identify organisms M, N and P. (3 mark)

M

N

P

(ii) All the three organisms above belong to the same kingdom. Name the kingdom to which they belong. (1 mark)

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(iii) State two main observable features of the kingdom you have named in (ii) above.

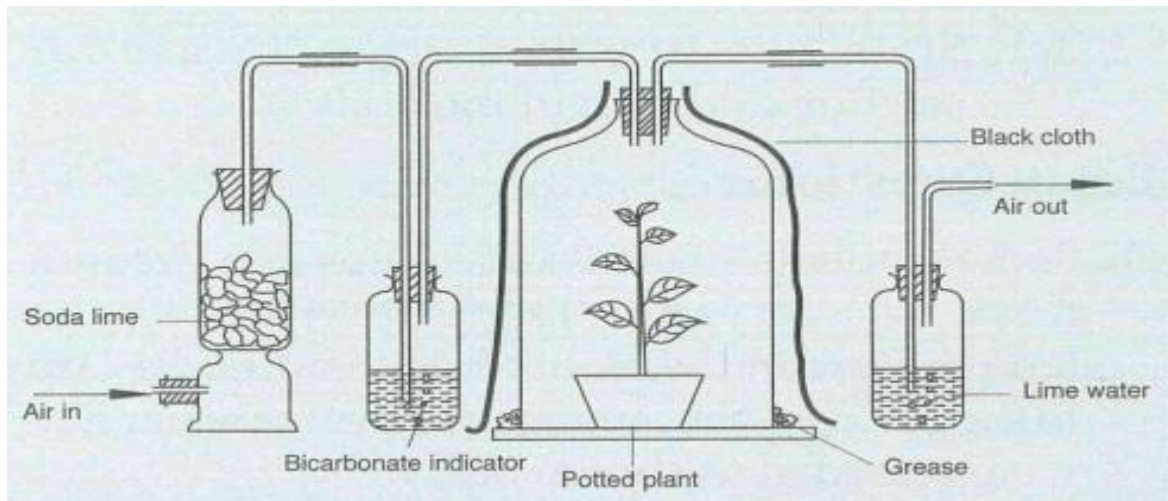
(2 marks)

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4. The apparatus below were used to investigate a certain physiological process in plants.



(a) State which physiological process was being investigated. (1 mark)

(b) Give the function of the following in the experiment:

(i) Soda lime (1 mark)

(ii) Bicarbonate indicator (1 mark)

(iii) Black cloth (1 mark)

(iv) Grease (1 mark)

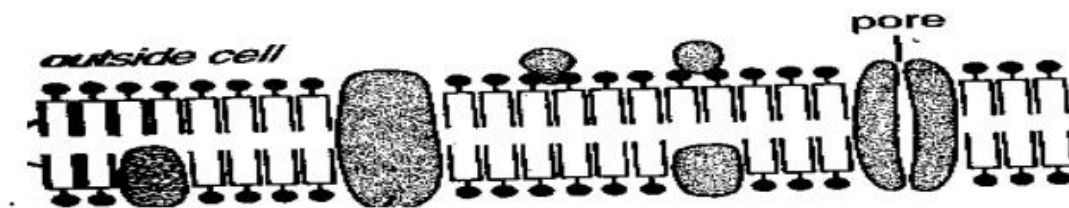
(c) (i) Suggest a possible observation that would be made at the end of this experiment.(1 mark)

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(ii) Give reasons for your answer in (c (i) above. (2 marks)

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5. Below is a diagram of a structure found in Eukaryotic cells. Study it and answer the questions that follow.



- (a) Identify the structure. (1 mark)
- (b) State two functions of the structure. (2 marks)
- (c) (i) Name one organelle found in animal cells but **absent** in plant cells. (1 mark)
- (ii) State one function of the organelle you have named in(c) above. (1 mark)
- (d) Briefly explain cell biology as evidence of evolution. (3 marks)

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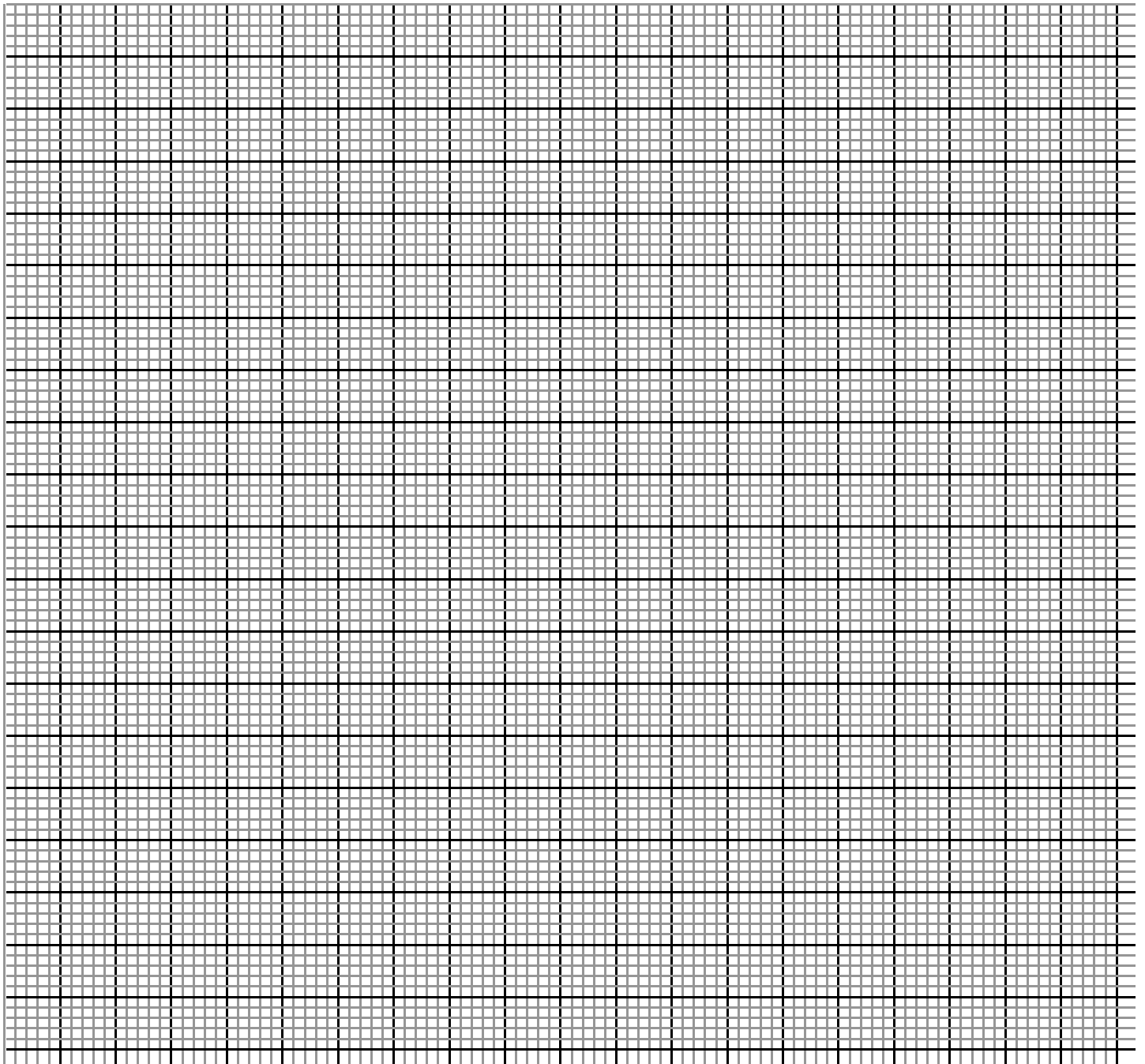
SECTION B (40 MARKS)

Answer question 6 (compulsory) and either question 7 or 8

6. The pressure in the flow of blood in a mammal was determined at two different vessels. The data was taken within a period of one minute and was presented as follows.

Time (seconds)	Blood vessel A	Blood vessel B
0	160	320
10	165	360
20	170	320
30	180	400
40	170	360
50	160	320
60	160	360

(a) Plot a graph of blood pressure in both vessels against time on the same axis. **(8 marks)**



(b) Describe the trend of each curve. **(2 marks)**

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(c) From the graphs, suggest the possible identity for:

(i) Blood vessel A. (1 mark)

(ii) Blood vessel B. (1 mark)

(d) Give reasons for your answer in (c) (i) and (ii) above. (2 marks)

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(e) Explain a factor that would result to an increase in blood pressure in both blood vessels above. (2 marks)

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(f) State three structural differences between the two vessels mentioned in (c) above. (3 marks)

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(g) Name a nerve that is responsible for the increase of the rate of heartbeat. (1 mark)

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7. (a) Describe the digestion of a starchy meal along the human alimentary canal. (13 marks)

(b) Describe the process of urea formation. (7 marks)

8. Discuss the effects of drugs abuse on human health. (20 marks)